

FDA Drug Recall Classifier

Predicting Recall Severity from Reason Text

Jannet Castaneda Sanchez

ID: 105863950

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BACKGROUND

Problem: FDA recall notices are manually classified into Class I (urgent) to Class III (least risk), often with a delay.

Goal: Predict recall severity from the "reason for recall" text using machine learning to support faster regulatory and clinical action.

Motivation:

- Class I recalls involve the risk of serious injury/death.
- Timely classification helps protect public health.

DATASET & ACQUISITION METHOD

Source: FDA Enforcement Report API: <https://api.fda.gov/drug/enforcement.json>

- Extracted ~15,000 drug recalls (2012–2024)
- Features used: reason_for_recall, classification, firm, product_description

Exploration Highlights:

- Class II dominates (60%), followed by Class III (30%) and Class I (10%)
 - Term Frequency used to identify key terms by class
 - Word clouds revealed distinct language patterns
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FRAMEWORK & MODEL

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TAKEAWAY & NEXT STEPS

Takeaways:

- Natural Language Processing-based classifier can assist in the early detection of urgent recalls
- Deployed end-to-end using API + Shiny App

Next Steps:

- Enable batch predictions
- Add model explainability

Repo: <https://github.com/jannet1313/fda-drug-recall-classifier>

Thank you!
